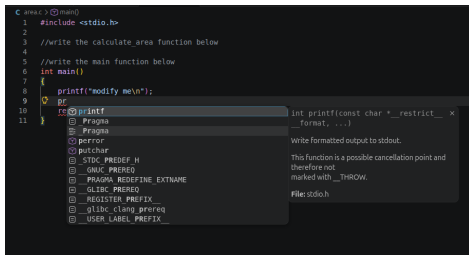


Lab 2: Control Flow and Time Complexity

Matteo Fasiolo

VSCode set up (ignore if you have already done this)

VSCode comes with auto-suggestion features:



These are useful for experienced programmers, but are detrimental for learners.

The exam is pencil-and-paper, so it's best to disable all these features.

The next slide explains how.

Open VSCode and press Ctrl+Shift+P (Cmd+Shift+P on Macs).

In the text box, type Preferences: Open User Settings (JSON).

Copy-paste this code (including brackets) into that file:

```
{
"chat.disableAIFeatures": true,
"editor.inlineSuggest.enabled": false,
"editor.quickSuggestions": false,
"editor.suggestOnTriggerCharacters": false,
"editor.wordBasedSuggestions": "off",
"editor.parameterHints.enabled": false,
"editor.snippetSuggestions": "none",
"editor.tabCompletion": "off",
"editor.acceptSuggestionOnEnter": "off",
"editor.acceptSuggestionOnCommitCharacter": false,
"C_Cpp.autocomplete": "disabled"
}
```

Lab set up

All course material will be on [Github](#).

Download lab2.zip from the Week_by_week/Week_2 folder and save it to a personal folder.

Extract the files.

Open Visual Studio Code, “open folder”, select the folder which contains the file you have just downloaded.

Homework 1: maximum of three integers

Open `max.c`

Write a function `max` at the specified place.

The function takes three integer inputs: `a`, `b`, `c`. It returns the maximum.

Your solution should use if-else statements.

Homework 2: Fibonacci sequence

The Fibonacci sequence starts with

$$F_0 = 0, \quad F_1 = 1,$$

and every later value is the sum of the previous two:

$$F_i = F_{i-1} + F_{i-2}.$$

For example, F_0 to F_{10} are

0 1 1 2 3 5 8 13 21 34 55

Open `fibonacci.c`.

Complete the program so that it prints $F_0, F_1, \dots, F_n$, where $n \geq 1$.

- Use a loop.
- Keep track of the previous two Fibonacci numbers.
- **Do not use recursion** (we have not discussed this yet).
- Test your program with $n = 10$ and check that its output matches the previous slide.

After your program works, answer the following questions:

1. What is the input size?
2. How many times does the loop run, in terms of n ?
3. What is the time complexity in Big-O notation?

Support your answer with one sentence about how the number of loop iterations grows when n grows.

Homework 3: counting handshakes

There are n people in a room, numbered $1, 2, \dots, n$.

Each pair of people shakes hands exactly once.

Open `handshakes.c` and complete the program using **nested loops**.

Print each pair and count the total number of handshakes.

For $n = 4$, the output should be

```
1 2
1 3
1 4
2 3
2 4
3 4
Total: 6
```

After your program works, answer the following questions:

1. What is the input size?
2. How many times does the inner loop body run, in terms of n ?
3. What is the time complexity in Big-O notation?
4. Approximately what happens to the amount of work when n doubles?

Submission

Upload your solutions to homeworks 2 and 3 in two separate .c files.

NOTE Blackboard allows uploading multiple files, so this should work fine.

Before submitting, make sure your code compiles and runs!